SCm CMB Angular Power Spectrum via Phonon-Modulated Primordial Integration

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Abstract

We derive the full SCm CMB angular power spectrum by integrating the phonon-modulated primordial power spectrum over k-space:

$$C_\ell^{\text{SCm}} = \frac{2}{\pi} \int dk\, k^2\, P_{\text{SCm}}(k)\, |\Delta_{T}^{\text{SCm}}(\ell, k)|^2$$

where the SCm-modified primordial spectrum is:

$$P_{\text{SCm}}(k) = A_s \left(\frac{k}{k_{\text{piv}}}\right)^{n_s - 1} \left[1 + \Phi \cdot S_{26} \cdot (2R - 1)\right]$$

This extends the simplified CMBAngularPowerCalculator (grok_100_equations_module.py) to a full k-space integration with Sachs-Wolfe, acoustic, and Silk damping transfer functions, all modulated by the SCm phonon linewidth Γ.

$\S$1 SCm-Modified Primordial Spectrum

The standard nearly scale-invariant spectrum $P(k) = A_s (k/k_{\text{piv}})^{n_s - 1}$ is boosted by the SCm buoyancy factor:

$$P_{\text{SCm}}(k) = P(k) \cdot \left[1 + \Phi_{1.25\,\text{THz}} \cdot S_{26}^{(3)}([\text{SSq}]) \cdot (2R - 1)\right]$$

with $A_s = 2.1 \times 10^{-9}$ (Planck 2018), $n_s = 0.965$, $k_{\text{piv}} = 0.05\;\text{Mpc}^{-1}$, $\Phi_0 = 10^{20}$, $S_{26} \approx 1.86$.

$\S$2 Temperature Transfer Function

$$\Delta_T^{\text{SCm}}(\ell, k) = \Delta_T^{\text{SW}}(\ell, k) + 0.6 \cdot \Delta_T^{\text{acoustic}}(\ell, k)$$

$\S$2.1 Sachs-Wolfe Term

$$\Delta_T^{\text{SW}} = \frac{1}{3} \exp\left(-\frac{x^2}{2}\right), \quad x = \frac{k \cdot r_*}{\ell + 0.5}$$

where $r_* \approx 14{,}000$ Mpc is the comoving sound horizon at last scattering.

$\S$2.2 Acoustic Oscillation Term

$$\Delta_T^{\text{acoustic}} = \cos(x) \cdot \exp\left(-\frac{k^2}{k_D^2}\right)$$

with Silk damping scale $k_D \approx 0.15\;\text{Mpc}^{-1}$.

$\S$3 Full Power Spectrum Integral

$$C_\ell^{\text{SCm}} = \frac{2}{\pi} \int_{k_{\min}}^{k_{\max}} dk\, k^2\, P_{\text{SCm}}(k)\, |\Delta_T^{\text{SCm}}(\ell, k)|^2 \cdot (T_0 \times 10^6)^2$$

in units of μK^2.

$\S$4 Acoustic Peak Structure

Peak	Multipole ℓ	C_ℓ^{SCm} (μK^2)	Physical Origin
1st	~ 220	Dominant	SW + first compression
2nd	~ 546	Subdominant	First rarefaction
3rd	~ 800	Tertiary	Second compression

The SCm boost factor $1 + \Phi S_{26} (2R-1)$ uniformly amplifies all peaks, preserving the observed peak ratio structure while providing a physical origin via vacuum phonon resonance.

$\S$5 Advantages over Standard Treatment

Aspect	Standard ΛCDM	SCm Phonon
Primordial spectrum	Inflaton slow-roll	SCm vacuum buoyancy
Free parameters	A_s, n_s, r	$\Phi_0, \Gamma, [\text{SSq}]$
Physical mechanism	Quantum fluctuations	Phonon resonance at 1.25 THz
Peak amplification	Static A_s	Dynamic $\Phi(\Gamma)$
Testability	CMB polarization	THz phonon linewidth

References

- PAPER_1093: SCm CMB Temperature Fluctuation
- PAPER_1094: CMB Buoyancy Sector Lagrangian
- PAPER_202: UQFF Reionization / BBN / Recombination
- PAPER_199: $F_{U,Bi,i}$ Taxonomy Part 2 — Cosmological Dark Sector

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